NOTES:

1. VENDOR SUPPLIED VALVES FIELD MOUNTED AS CLOSE AS POSSIBLE TO BEARING SEALS.
2. CONNECTIONS INDICATED ARE VENDOR TERMINATIONS. MATING FITTINGS TO BE SUPPLIED BY THE PIPING CONTRACTOR U.N.O.
3. DIVIDER TO BE LOCATED IN CLOSE PROXIMITY TO TRUNNION BEARING HOUSE.
4. CALCULATE THE DIFFERENCE BETWEEN THE HIGHEST AND LOWEST TEMPERATURE OF THE THREE SIGNALS.
5. REFER TO VENDOR P&ID: 2000-ME001.3-D07-50002 FOR DETAILS.
6. FOR VALVE BANK DETAILS REFER TO 2000-F13-PID-PR-10013.

MILL AIR RECEIVER
2000-F24-PID-PR-10011
PLANT AIR

13-2633-IA-MSG3-25

COOLING WATER HEADER
2000-F13-PID-PR-10009
COOLING WATER

13-0114-COW-MSG1-100-TI

FE 13413 · FIT 13413 · FI 13413 L

DETAIL 1
TCV 13416

100VBFD1L2 F13-10322
100VBFD1L2 F13-10323
25VB03S1 F13-10326
TO GRADE
F0
100VBFD1L2 F13-10325
100VCD1C1 F13-10328
25VB03S1 F13-10327
TO GRADE
100VBFD1L2 F13-10324
13-1020-COW-MSG1-100-TI

COOLING WATER HEADER
2000-F13-PID-PR-10009
COOLING WATER

13-0146-COW-MSG1-100-TI
100VBFD1L2 F13-10319
25VB03S1 F13-10320
TO GRADE
100VCD1C1 100x80 F13-10321

13-0111-COW-MSG1-100
100x80
13-1364-LO-SS3-25
13-1365-LO-SS3-25
25VB03S1 F13-10330
TO GRADE

PINION GEAR FLANK TEMPERATURE (RIGHT SIDE)
TI 13763 · TI 13764 · TI 13765 · TI 13766 · TDI 13763
15-MECH-MSG3

SP HS-053 · SP HS-053
TI 13955 · VI 13972 · TI 13957 · VI 13974
SP HS-049 4 OFF
13-2354-LO-SS1-80
13-1378-LO-SS1-80

100VBFD1L2 F13-10329
13-2072-COW-MSG1-100
100VBFD1L2 F13-10371

13-0111-COW-MSG1-100
COOLING WATER RETURN HEADER
2000-F13-PID-PR-10009
COOLING WATER RETURN

F13-FN-453
BALL MILL LUBE ROOM FAN (INDUCTION)
F13-FN-454
BALL MILL LUBE ROOM FAN (EXTRACTION)

FIXED FLOATING PINION BEARING LUBRICATION
FI 13911 · FI 13910 · FI 13909 · FI 13908

LUBE OIL RESERVOIR
TI 13866 · LI 13874 · TI 13873 · LI 13872

F13-HC-451
F13-HD-451
BALL MILL LUBE ROOM MAINTENANCE MONORAIL AND TROLLEY

LUBE OIL DISTRIBUTION
ZI 13880 · ZSC 13880 · ZSO 13880 · PDI 13879 · TIC 13880 · FI 13881

HP LUBE PUMPS
SAL 13865 · SAL 13864 · SAL 13863 · FI 13891

THRUST RAIL LUBE
FI 13892

BALL MILL MAIN BEARINGS LUBRICATION CONTROL PANEL

HS 13875D	HS 13875C	HS 13877B	HS 13877C	HS 13903B	HS 13903C	HS 13905B	HS 13905C	HS 13885B	HS 13885C	HS 13887B	
HS 13887C	HS 13889B	HS 13889C	HS 13925A	HS 13925B	HS 13925C	HS 13968A	HS 13968B	HS 13968C	XI 13914	XI 13915	
HS 13913	HS 13953	XI 13912	XI 13937	XI 13940	XI 13943	XI 13946	XI 13949	XI 13952	XI 13976	XI 13916	
XI 13917	XI 13918	XI 13920	XI 13921	XI 13922	XI 13923	XI 13919	XI 13924	XI 13926	XI 13927	XI 13928	
XI 13929	XI 13930	XI 13931	XI 13932	XI 13933	XI 13934	XI 13935	XI 13936	XI 13941	XI 13942	XI 13951	
XI 13948	XI 13950	XI 13947									

F13-LU-423
BALL MILL MAIN BEARINGS LUBRICATION UNIT

13-1354-LO-SS9-38 FEED
13-1355-LO-SS9-38 FEED
13-1356-LO-SS9-38 FEED
13-1357-LO-SS9-38 FEED
VALVE BANK TYPE 1

13-1358-LO-SS9-38 THRUST
13-1359-LO-SS9-38 THRUST
13-1360-LO-SS9-38 THRUST
13-1361-LO-SS9-38 THRUST
VALVE BANK TYPE 2

13-1350-LO-SS9-38
13-1351-LO-SS9-38 DELIVERY
13-1352-LO-SS9-38 DELIVERY
13-1353-LO-SS9-38 DELIVERY
VALVE BANK TYPE 1
NOTE 6

SP HS-047 4 OFF · SP HS-048 4 OFF · SP HS-047 4 OFF
SP SS-033 · SP SS-034

DELIVERY END
F13-MB-401
BALL MILL
NOTE 5
FEED END

OCM UNIT
START XC 13201A
STOP XC 13201B
ALARM XC 13201

RETURN DN200
13-1362-LO-SS1-200 10°
RETURN

KI 13494 · KI 13493 · KI 13491 · KI 13492
SP HS-020 · SP HS-019 · 20x12
25VB03S1 F13-14236

13-1077-LO-SS8-16-ET
13-1072-LO-SS8-16-ET
13-1496-IA-MSG3-25
13-1440-LO-SS8-20-ET
13-1249-LO-SS8-16-ET
13-1250-LO-SS8-16-ET
ET

BALL MILL SEAL GREASE SYSTEM 2000-F13-PID-PR-10013 GREASE
BALL MILL SEAL GREASE SYSTEM 2000-F13-PID-PR-10013 GREASE
MILLS RING GEAR SPRAY SYSTEM 2000-F13-PID-PR-10013 AIR
MILLS RING GEAR SPRAY SYSTEM 2000-F13-PID-PR-10013 GREASE
BALL MILL SEAL GREASE SYSTEM 2000-F13-PID-PR-10013 GREASE
BALL MILL SEAL GREASE SYSTEM 2000-F13-PID-PR-10013 GREASE

PINION GEAR FLANK TEMPERATURE (LEFT SIDE)
TI 13954 · TI 13956 · VI 13971 · VI 13973 · TDI 13786 · TI 13783 · TI 13784 · TI 13785 · TI 13786
15-MECH-MSG3

13-1366-LO-SS3-25
13-1367-LO-SS3-25
SP HS-053 2 OFF
SP HS-049 4 OFF
13-1379-LO-SS1-80
13-2355-LO-SS1-80

13-1363-LO-SS1-200

GIRTH GEAR GUARD
13-0136-FD-MS1-100
WASTE SKIP

25VB03S1 F13-14245
SP MS-001
13-2073-COW-MSG1-100

MILL AIR RECEIVER
2000-F13-PID-PR-10010
PLANT AIR

13-2631-IA-MSG3-25
FI 13415 L
FIT 13415 · FE 13415

13-0115-COW-MSG1-100-TI
COOLING WATER HEADER
2000-F13-PID-PR-10009
COOLING WATER

100VBFD1L2 F13-10312
100VBFD1L2 F13-10313
25VB03S1 F13-10316
TO GRADE
100VBFD1L2 F13-10314
13-0310-COW-MSG1-100-TI
100VBFD1L2 F13-10315
25VB03S1 F13-10317
TO GRADE

DETAIL 1
TCV 13416
13-1363-LO-SS1-200 10°

MOTOR BEARING RTD (NDE) VI 13767
MOTOR BEARING RTD (DE) VI 13770
TIC 13414 · TIT 13414 · TE 13414
DN100

F13-MB-401A
BALL MILL MOTOR 1

80x65
13-1372-LO-SS1-80
13-1373-LO-SS1-80
25VB03S1 F13-14177
TO GRADE
13-1376-LO-SS1-80 10°

13-1368-LO-SS8-20
13-1369-LO-SS8-20

13-0138-COW-MSG1-100
25VB03S1 F13-10332
100VBFD1L2 F13-10331
TO GRADE
COOLING WATER RETURN HEADER
2000-F13-PID-PR-10009
COOLING WATER RETURN

FEED TRUNNION BEARING RTDS (RIGHT SIDE)
NOTE 4
TDI 13964 f(x)
TI 13964A OUTBOARD
TI 13964B CENTRE
TI 13964C INBOARD

DISCHARGE TRUNNION BEARING RTDS (RIGHT SIDE)
NOTE 4
TDI 13958 f(x)
TI 13958A OUTBOARD
TI 13958B CENTRE
TI 13958C INBOARD

THRUST BEARING (RIGHT SIDE)
TI 13962A OUTBOARD
TI 13962B INBOARD

FEED TRUNNION BEARING RTDS (LEFT SIDE)
NOTE 4
TDI 13966 f(x)
TI 13966A OUTBOARD
TI 13966B CENTRE
TI 13966C INBOARD

DISCHARGE TRUNNION BEARING RTDS (LEFT SIDE)
NOTE 4
TDI 13960 f(x)
TI 13960A OUTBOARD
TI 13960B CENTRE
TI 13960C INBOARD

THRUST BEARING (LEFT SIDE)
TI 13967A OUTBOARD
TI 13967B INBOARD

LUBE OIL RESERVOIR
LAL 13819 · LALL 13819
TSH 13820 · TSL 13820
PDAH 13816 · PDAHH 13816
PDAH 13817 · PDAHH 13817
TI 13821
LI 13822

MOTOR 2 DE/NDE LUBE OIL SUPPLY
FI 13829 · FI 13830 · PAL 13825 · PI 13825 · PAL 13826 · PI 13826

MOTOR 1 DE/NDE LUBE OIL SUPPLY
FI 13827 · FI 13828 · PAL 13823 · PI 13823 · PAL 13824 · PI 13824

F13-LU-421
BALL MILL MOTORS LUBRICATION UNIT
80x40

MOTOR BEARING RTD (NDE) VI 13789
MOTOR BEARING RTD (DE) VI 13792
TIC 13416 · TIT 13416 · TE 13416

100VCD1C1 F13-10318
F13-MB-401B
BALL MILL MOTOR 2
25VB03S1 F13-14178
TO GRADE
80x65 · 80x65
13-1374-LO-SS1-80
13-1375-LO-SS1-80
13-1377-LO-SS1-80 10°

DN100
13-0139-COW-MSG1-100
13-1370-LO-SS8-20
13-1371-LO-SS8-20
25VB03S1 F13-10334
100VBFD1L2 F13-10333
TO GRADE
COOLING WATER RETURN HEADER
2000-F13-PID-PR-10009
COOLING WATER RETURN

100VBFD1L2 F13-10372